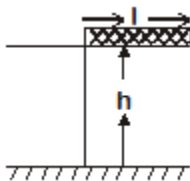




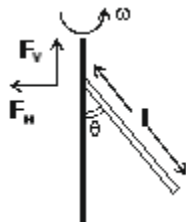
Unit-1 (Mechanics)

1. A rectangular solid box of length 0.3 m is held horizontally, with one of its sides on the edge of a platform of height 5 m. When released, it slips off the table in a very short time $\tau = 0.01$ s, remaining essentially horizontal. The angle by which it would rotate when it hits the ground will be (in radians) close to:



- (1) 0.3 (2) 0.02
(3) 0.28 (4) 0.5
2. A person of mass M is, sitting on a swing of length L and swinging with an angular amplitude θ_0 . If the person stands up when the swing passes through its lowest point, the work done by him, assuming that his centre of mass moves by a distance l ($l < L$), is close to :
- (1) Mgl (2) $Mgl(1 + \theta_0^2)$
(3) $Mgl\left(1 + \frac{\theta_0^2}{2}\right)$ (4) $Mgl(1 - \theta_0^2)$
3. In a car race on straight road, car A takes a time t less than car B at the finish and passes finishing point with a speed ' v ' more than that of car B . Both the cars start from rest and travel with constant acceleration a_1 and a_2 respectively. Then ' v ' is equal to:
- (1) $\frac{a_1 + a_2}{2} t$ (2) $\frac{2a_1 a_2}{a_1 + a_2} t$
(3) $\sqrt{2a_1 a_2} t$ (4) $\sqrt{a_1 a_2} t$
4. A uniform rod of length ' l ' is pivoted at one of its ends on a vertical shaft of negligible radius. When the shaft rotates at angular speed ω the rod makes an angle θ with it (see figure). To find θ equate the rate of change

of angular momentum (direction going into the paper) $\frac{ml^2}{12} \omega^2 \sin\theta \cos\theta$ about the centre of mass (CM) to the torque provided by the horizontal and vertical forces F_H and F_V about the CM. The value of θ is then such that



(1) $\cos\theta = \frac{g}{l\omega^2}$

(2) $\cos\theta = \frac{2g}{3l\omega^2}$

(3) $\cos\theta = \frac{g}{2l\omega^2}$

(4) $\cos\theta = \frac{3g}{2l\omega^2}$

5. Moment of inertia of a cylinder of mass M , length L and radius R about an axis passing through its centre and perpendicular to the axis of the cylinder is $I = M \left(\frac{R^2}{4} + \frac{L^2}{12} \right)$. If such a cylinder is to be made for a given mass of a material, the ratio L/R for it to have minimum possible I is

(1) $\sqrt{\frac{2}{3}}$

(2) $\frac{2}{3}$

(3) $\frac{3}{2}$

(4) $\sqrt{\frac{3}{2}}$

6. If R_E be the radius of Earth, then the ratio between the acceleration due to gravity at a depth ' r ' below and a height ' r ' above the earth surface is:
(Given : $r < R_E$)

(1) $1 - \frac{r}{R_E} - \frac{r^2}{R_E^2} - \frac{r^3}{R_E^3}$

(2) $1 + \frac{r}{R_E} + \frac{r^2}{R_E^2} + \frac{r^3}{R_E^3}$

(3) $1 + \frac{r}{R_E} - \frac{r^2}{R_E^2} - \frac{r^3}{R_E^3}$

(4) $1 + \frac{r}{R_E} - \frac{r^2}{R_E^2} + \frac{r^3}{R_E^3}$

7. If Y , K and η are the values of Young's modulus, bulk modulus and modulus of rigidity of any material respectively. Choose the correct relation for these parameters.

(1) $Y = \frac{9K\eta}{3K - \eta} \text{ N/m}^2$

(2) $Y = \frac{9K\eta}{2\eta + 3K} \text{ N/m}^2$

(3) $K = \frac{Y\eta}{9\eta - 3Y} \text{ N/m}^2$

(4) $\eta = \frac{3YK}{9K + Y} \text{ N/m}^2$

8. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Body 'P' having mass M moving with speed ' u ' has head-on collision elastically with another body 'Q' having mass ' m ' initially at rest. If $m \ll M$, body 'Q' will have a maximum speed equal to ' $2u$ ' after collision.

Reason R : During elastic collision, the momentum and kinetic energy are both conserved.

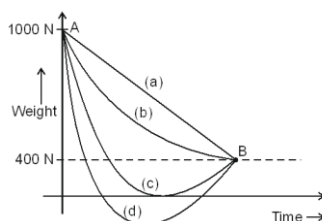
In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both A and R are correct and R is the correct explanation of A
- (2) A is not correct but R is correct
- (3) A is correct but R is not correct
- (4) Both A and R are correct but R is NOT the correct explanation of A

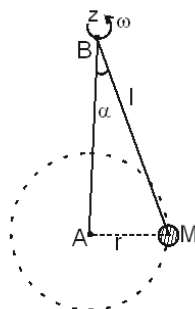
9. A boy reaches the airport and finds that the escalator is not working. He walks up the stationary escalator in time t_1 . If he remains stationary on a moving escalator then the escalator takes him up in time t_2 . The time taken by him to walk up on the moving escalator will be:

- (1) $t_2 - t_1$
- (2) $\frac{t_1 t_2}{t_2 - t_1}$
- (3) $\frac{t_1 + t_2}{2}$
- (4) $\frac{t_1 t_2}{t_2 + t_1}$

10. A person whose mass is 100 kg travels from Earth to Mars in a spaceship. Neglect all other objects in sky and take acceleration due to gravity on the surface of the Earth and Mars as 10 m/s^2 and 4 m/s^2 respectively. Identify from the below figures, the curve that fits best for the weight of the passenger as a function of time.



- (1) (a)
 - (2) (c)
 - (3) (d)
 - (4) (b)
11. A mass M hangs on a massless rod of length l which rotates at a constant angular frequency. The mass M moves with steady speed in a circular path of constant radius. Assume that the system is in steady circular motion with constant angular velocity ω . The angular momentum of M about point A is L_A which lies in the positive z direction and the angular momentum of M about point B is L_B . The correct statement for this system is :



- (1) L_A is constant, both in magnitude and direction
 - (2) L_B is constant in direction with varying magnitude
 - (3) L_A and L_B are both constant in magnitude and direction
 - (4) L_B is constant, both in magnitude and direction
12. In the experiment of Ohm's law, a potential difference of 5.0 V is applied across the end of a conductor of length 10.0 cm and diameter of 5.00 mm. The measured current in the conductor is 2.00 A. The maximum permissible percentage error in the resistivity of the conductor is:
- (1) 3.0
 - (2) 3.9
 - (3) 7.5
 - (4) 8.4
13. Three blocks A, B and C are lying on a smooth horizontal surface, as shown in the figure. A and B have equal masses, m while C has mass M . Block A is given an initial speed v towards B due to which it collides with B perfectly inelastically. The combined mass collides with C, also perfectly inelastically. $\frac{5}{6}$ of the initial kinetic energy is lost in whole process. What is value of M/m ?
-
- (1) 3
 - (2) 4
 - (3) 2
 - (4) 5
14. A particle of mass m is dropped from a height h above the ground. At the same time another particle of the same mass is thrown vertically upwards from the ground with a speed of $\sqrt{2gh}$. If they collide head-on completely inelastically, the time taken for the combined mass to reach the ground, in units of $\sqrt{\frac{h}{g}}$ is
- (1) $\sqrt{\frac{1}{2}}$
 - (2) $\sqrt{\frac{3}{4}}$
 - (3) $\frac{1}{2}$
 - (4) $\frac{1}{2}$



15. A particle of mass m is projected with a speed u from the ground at an angle $\theta = \frac{\pi}{3}$ w.r.t. horizontal (x-axis).

When it has reached its maximum height, it collides completely inelastically with another particle of the same mass and velocity $u\hat{i}$. The horizontal distance covered by the combined mass before reaching the ground is

- (1) $\frac{5u^2}{8g}$ (2) $\frac{3\sqrt{2}u^2}{4g}$
 (3) $\frac{3\sqrt{3}u^2}{8g}$ (4) $2\sqrt{2}\frac{u^2}{g}$

16. Mass per unit area of a circular disc of radius a depends on the distance r from its centre as $\sigma(r) = A + Br$. The moment of inertia of the disc about the axis, perpendicular to the plane and passing through its centre is:

- (1) $2\pi a^4 \left(\frac{A}{4} + \frac{B}{5} \right)$ (2) $\pi a^4 \left(\frac{A}{4} + \frac{aB}{5} \right)$
 (3) $2\pi a^4 \left(\frac{A}{4} + \frac{aB}{5} \right)$ (4) $2\pi a^4 \left(\frac{aA}{4} + \frac{B}{5} \right)$

17. A stellite of mass m is launched vertically upwards with an initial speed u from the surface of the earth. After it reaches height R (R = radius of the earth), it ejects a rocket of mass $\frac{m}{10}$ so that subsequently the satellite moves in a circular orbit. The kinetic energy of the rocket is (G is the gravitational constant; M is the mass of the earth):

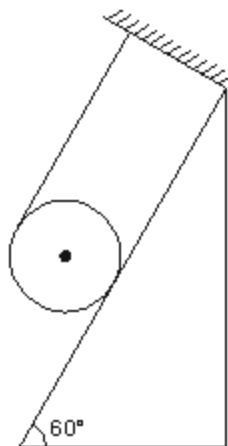
- (1) $5m \left(u^2 - \frac{119GM}{200R} \right)$ (2) $\frac{3m}{8} \left(u + \sqrt{\frac{5GM}{6R}} \right)^2$
 (3) $\frac{m}{20} \left(u - \sqrt{\frac{2GM}{3R}} \right)^2$ (4) $\frac{m}{20} \left(u^2 + \frac{113GM}{200R} \right)$

18. A rubber ball is released from a height of 5 m above the floor. It bounces back repeatedly, always rising to $\frac{81}{100}$ of the height through which it falls. Find the average speed of the ball.

(Take $g = 10 \text{ ms}^{-2}$)

- (1) 3.50 ms^{-1} (2) 2.0 ms^{-1}
 (3) 2.50 ms^{-1} (4) 3.0 ms^{-1}

19. A solid cylinder of mass m is wrapped with an inextensible light string and, is placed on a rough inclined plane as shown in the figure. The frictional force acting between the cylinder and the inclined plane is:



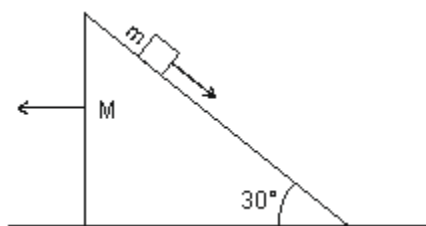
[The coefficient of static friction, μ_s , is 0.4]

- (1) 0 (2) $5 mg$
(3) $\frac{7}{2} mg$ (4) $\frac{mg}{5}$

20. A block of mass m slides on the wooden wedge, which in turn slides backward on the horizontal surface. The acceleration of the block with respect to the wedge is:

Given $m = 8 \text{ kg}$, $M = 16 \text{ kg}$

Assume all the surfaces shown in the figure to be frictionless.



- (1) $\frac{2}{3} g$ (2) $\frac{4}{3} g$
(3) $\frac{6}{5} g$ (4) $\frac{3}{5} g$

21. A body is projected vertically upwards from the surface of earth with a velocity sufficient enough to carry it to infinity. The time taken by it to reach height h is _____s.

- (1) $\sqrt{\frac{2R_e}{g}} \left[\left(1 + \frac{h}{R_e} \right)^{\frac{3}{2}} - 1 \right]$ (2) $\frac{1}{3} \sqrt{\frac{R_e}{2g}} \left[\left(1 + \frac{h}{R_e} \right)^{\frac{3}{2}} - 1 \right]$
(3) $\sqrt{\frac{R_e}{2g}} \left[\left(1 + \frac{h}{R_e} \right)^{\frac{3}{2}} - 1 \right]$ (4) $\frac{1}{3} \sqrt{\frac{2R_e}{g}} \left[\left(1 + \frac{h}{R_e} \right)^{\frac{3}{2}} - 1 \right]$



22. Water droplets are coming from an open tap at a particular rate. The spacing between a droplet observed at 4th second after its fall to the next droplet is 34.3 m. At what rate the droplets are coming from the tap? (Take $g = 9.8 \text{ m/s}^2$)
- (1) 1 drop / 7 seconds (2) 3 drops / 2 seconds
(3) 2 drops / second (4) 1 drop / second
23. A body of mass M moving at speed V_0 collides elastically with a mass ' m ' at rest. After the collision, the two masses move at angles θ_1 and θ_2 with respect to the initial direction of motion of the body of mass M . The largest possible value of the ratio M/m , for which the angles θ_1 and θ_2 will be equal, is:
- (1) 4 (2) 2
(3) 1 (4) 3
24. Suppose you have taken a dilute solution of oleic acid in such a way that its concentration becomes 0.01 cm^3 of oleic acid per cm^3 of the solution. Then you make a thin film of this solution (monomolecular thickness) of area 4 cm^2 by considering 100 spherical drops of radius $\left(\frac{3}{40\pi}\right)^{\frac{1}{3}} \times 10^{-3} \text{ cm}$. Then the thickness of oleic acid layer will be $x \times 10^{-14} \text{ m}$. Where x is _____.

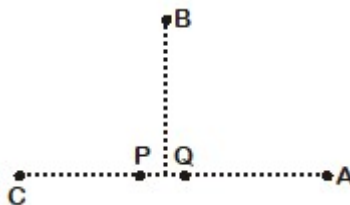
ANSWER KEY UNIT - 1

- | | | | | |
|---------|---------|---------|----------|---------|
| 1. (4) | 2. (2) | 3. (4) | 4. (4) | 5. (4) |
| 6. (3) | 7. (3) | 8. (1) | 9. (4) | 10. (2) |
| 11. (1) | 12. (2) | 13. (2) | 14. (4) | 15. (3) |
| 16. (3) | 17. (1) | 18. (3) | 19. (4) | 20. (1) |
| 21. (4) | 22. (4) | 23. (4) | 24. (25) | |



Unit-2 (Oscillations & Wave)

- * 1. A damped harmonic oscillator has a frequency of 5 oscillations per second. The amplitude drops to half its value for every 10 oscillations. The time it will take to drop to $\frac{1}{1000}$ of the original amplitude is close to:
- (1) 100 s (2) 10 s
(3) 50 s (4) 20 s
2. A simple pendulum of length 1 m is oscillating with an angular frequency 10 rad/s. The support of the pendulum starts oscillating up and down with a small angular frequency of 1 rad/s and an amplitude of 10^{-2} m. The relative change in the angular frequency of the pendulum is best given by
- (1) 10^{-3} rad/s (2) 10^{-1} rad/s
(3) 10^{-5} rad/s (4) 1 rad/s
3. Assume that the displacement (s) of air is proportional to the pressure difference (Δp) created by a sound wave. Displacement (s) further depends on the speed of sound (v), density of air (ρ) and the frequency (f). If $\Delta p \sim 10$ Pa, $v \sim 300$ m/s, $\rho \sim 1$ kg/m³ and $f \sim 1000$ Hz, then s will be of the order of (take the multiplicative constant to be 1)
- (1) $\frac{3}{100}$ mm (2) 10 mm
(3) 1 mm (4) $\frac{1}{10}$ mm
4. In the figure below, P and Q are two equally intense coherent sources emitting radiation of wavelength 20 m. The separation between P and Q is 5 m and the phase of P is ahead of that of Q by 90° . A, B and C are three distinct points of observation, each equidistant from the midpoint of PQ. The intensities of radiation at A, B, C will be in the ratio



(1) $0 : 1 : 2$

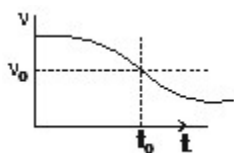
(2) $4 : 1 : 0$

(3) $2 : 1 : 0$

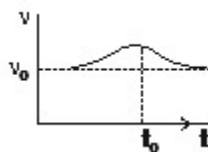
(4) $0 : 1 : 4$

- * 5. A sound source S is moving along a straight track with speed v , and is emitting sound of frequency v_0 (see figure). An observer is standing at a finite distance, at the point O, from the track. The time variation of frequency heard by the observer is best represented by

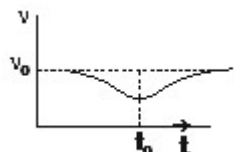
(t_0 represents the instant when the distance between the source and observer is minimum)



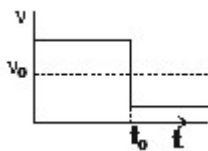
(1)



(2)

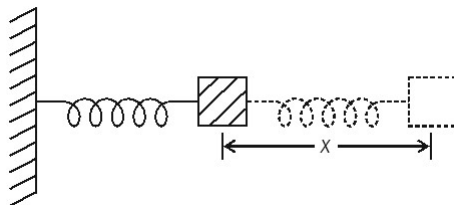


(3)



(4)

6. The motion of a mass on a spring, with spring constant K is as shown in figure.



The equation of motion is given by $x(t) = A \sin \omega t + B \cos \omega t$ with

Suppose that at time $t = 0$, the position of mass is $x(0)$ and velocity $v(0)$, then its displacement can also be represented as $x(t) = C \cos(\omega t - \phi)$, where C and ϕ are:

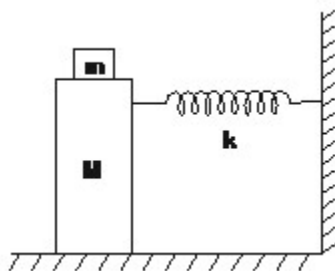
(1) $C = \sqrt{\frac{2v(0)^2}{\omega^2} + x(0)^2}, \phi = \tan^{-1}\left(\frac{x(0)\omega}{2v(0)}\right)$

(2) $C = \sqrt{\frac{v(0)^2}{\omega^2} + x(0)^2}, \phi = \tan^{-1}\left(\frac{x(0)\omega}{v(0)}\right)$

(3) $C = \sqrt{\frac{v(0)^2}{\omega^2} + x(0)^2}, \phi = \tan^{-1}\left(\frac{v(0)}{x(0)\omega}\right)$

(4) $C = \sqrt{\frac{2v(0)^2}{\omega^2} + x(0)^2}, \phi = \tan^{-1}\left(\frac{v(0)}{x(0)\omega}\right)$

7. In the given figure, a mass M is attached to a horizontal spring which is fixed on one side to a rigid support. The spring constant of the spring is k . The mass oscillates on a frictionless surface with time period T and amplitude A . When the mass is in equilibrium position, as shown in the figure, another mass m is gently fixed upon it. The new amplitude of oscillation will be :



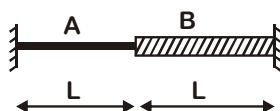
(1) $A\sqrt{\frac{M-m}{M}}$

(2) $A\sqrt{\frac{M}{M-m}}$

(3) $A\sqrt{\frac{M}{M+m}}$

(4) $A\sqrt{\frac{m}{M+m}}$

8. A wire of length $2L$, is made by joining two wires A and B of same length but different radii r and $2r$ and made of the same material. It is vibrating at a frequency such that the joint of the two wires forms a node. If the number of antinodes in wire A is p and that in B is q then the ratio $p : q$ is



(1) 4 : 9

(2) 1 : 2

(3) 3 : 5

(4) 1 : 4

9. A rod of mass ' M ' and length ' $2L$ ' is suspended at its middle by a wire. It exhibits torsional oscillations; if two masses each of ' m ' are attached at distance ' $L/2$ ' from its centre on both sides, it reduces the oscillation frequency by 20%. The value of ratio m/M is close to:

(1) 0.77

(2) 0.17

(3) 0.37

(4) 0.57

10. A heavy ball of mass M is suspended from the ceiling of a car by a light string of mass m ($m \ll M$). When the car is at rest, the speed of transverse waves in the string is 60 ms^{-1} . When the car has acceleration a , the wave-speed increases to 60.5 ms^{-1} . The value of a , in terms of gravitational acceleration g , is closest to

(1) $\frac{g}{30}$

(2) $\frac{g}{5}$

(3) $\frac{g}{20}$

(4) $\frac{g}{10}$

11. A cylindrical plastic bottle of negligible mass is filled with 310 ml of water and left floating in a pond with still water. If pressed downward slightly and released, it starts performing simple harmonic motion at angular frequency ω . If the radius of the bottle is 2.5 cm then ω is close to: (density of water = 10^3 kg/m^3)

(1) 2.50 rad s^{-1}

(2) 3.75 rad s^{-1}

(3) 5.00 rad s^{-1}

(4) 7.91 rad s^{-1}

12. A closed organ pipe has a fundamental frequency of 1.5 kHz. The number of overtones that can be distinctly heard by a person with this organ pipe will be : (Assume that the highest frequency a person can hear is 20,000 Hz)
- (1) 7 (2) 4
(3) 6 (2) 5
13. A string of length 1 m and mass 5 g is fixed at both ends. The tension in the string is 8.0 N. The string is set into vibration using an external vibrator of frequency 100 Hz. The separation between successive nodes on the string is close to
- (1) 33.3 cm (2) 10.0 cm
(3) 16.6 cm (4) 20.0 cm
14. Equation of travelling wave on a stretched string of linear density 5 g/m is $y = 0.03 \sin(450t - 9x)$ where distance and time are measured in SI units. The tension in the string is
- (1) 10 N (2) 7.5 N
(3) 5 N (4) 12.5 N
15. A resonance tube is old and has jagged end. It is still used in the laboratory to determine velocity of sound in air. A tuning fork of frequency 512 Hz produces first resonance when the tube is filled with water to a mark 11 cm below a reference mark, near the open end of the tube. The experiment is repeated with another fork of frequency 256 Hz which produces first resonance when water reaches a mark 27 cm below the reference mark. The velocity of sound in air, obtained in the experiment, is close to
- (1) 322 ms^{-1} (2) 341 ms^{-1}
(3) 328 ms^{-1} (4) 335 ms^{-1}



ANSWER KEY

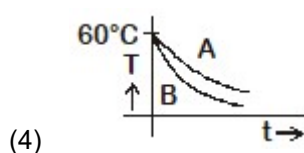
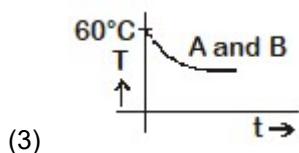
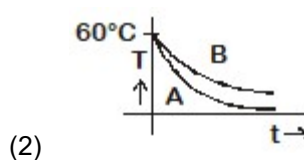
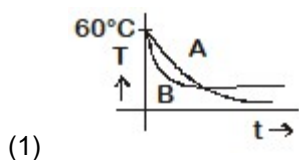
UNIT-2

- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (4) | 2. (1) | 3. (1) | 4. (3) | 5. (1) |
| 6. (3) | 7. (3) | 8. (2) | 9. (3) | 10. (2) |
| 11. (4) | 12. (3) | 13. (4) | 14. (4) | 15. (3) |



Unit-3 (Heat and Thermodynamics)

- * 1. Two identical beakers A and B contain equal volumes of two different liquids at 60°C each and left to cool down. Liquid in A has density of $8 \times 10^2 \text{ kg/m}^3$ and specific heat of $2000 \text{ Jkg}^{-1}\text{K}^{-1}$ while liquid in B has density of 10^3 kgm^{-3} and specific heat of $4000 \text{ Jkg}^{-1}\text{K}^{-1}$. Which of the following best describes their temperature versus time graph schematically? (Assume the emissivity of both the beakers to be the same)



2. A $25 \times 10^{-3} \text{ m}^3$ volume cylinder is filled with 1 mol of O_2 gas at room temperature (300 K). The molecular diameter of O_2 , and its root mean square speed, are found to be 0.3 nm and 200 m/s, respectively. What is the average collision rate (per second) for an O_2 molecule?

(1) $\sim 10^{12}$

(2) $\sim 10^{10}$

(3) $\sim 10^{11}$

(4) $\sim 10^{13}$

3. The number density of molecules of a gas depends on their distance r from the origin as, $n(r) = n_0 e^{-\alpha r^4}$. Then the total number of molecules is proportional to

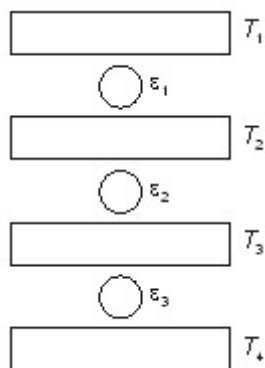
(1) $n_0 \alpha^{-3/4}$

(2) $\sqrt{n_0} \alpha^{1/2}$

(3) $n_0 \alpha^{1/4}$

(4) $n_0 \alpha^{-3}$

- ** 4. Three Carnot engines operate in series between a heat source at a temperature T_1 and a heat sink at temperature T_4 (see figure). There are two other reservoirs at temperature T_2 and T_3 , as shown, with $T_1 > T_2 > T_3 > T_4$. The three engines are equally efficient if



(1) $T_2 = (T_1 T_4)^{1/3}$; $T_3 = (T_1^2 T_4)^{1/3}$

(2) $T_2 = (T_1^3 T_4)^{1/4}$; $T_3 = (T_1 T_4^3)^{1/4}$

(3) $T_2 = (T_1 T_4)^{1/2}$; $T_3 = (T_1^2 T_4)^{1/3}$

(4) $T_2 = (T_1^2 T_4)^{1/3}$; $T_3 = (T_1 T_4^2)^{1/3}$

5. A vertical closed cylinder is separated into two parts by a frictionless piston of mass m and of negligible thickness. The piston is free to move along the length of the cylinder. The length of the cylinder above the piston is l_1 , and that below the piston is l_2 , such that $l_1 > l_2$. Each part of the cylinder contains n moles of an ideal gas at equal temperature T . If the piston is stationary, its mass, m will be given by :

(R is universal gas constant and g is the acceleration due to gravity)

(1) $\frac{nRT}{g} \left[\frac{l_1 - l_2}{l_1 l_2} \right]$

(2) $\frac{nRT}{g} \left[\frac{1}{l_2} + \frac{1}{l_1} \right]$

(3) $\frac{RT}{g} \left[\frac{2l_1 + l_2}{l_1 l_2} \right]$

(4) $\frac{RT}{ng} \left[\frac{l_1 - 3l_2}{l_1 l_2} \right]$

6. An ideal gas is enclosed in a cylinder at pressure of 2 atm and temperature, 300 K. The mean time between two successive collisions is 6×10^{-8} s. If the pressure is doubled and temperature is increased to 500 K, the mean time between two successive collisions will be close to

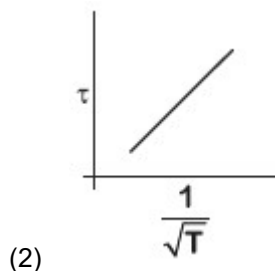
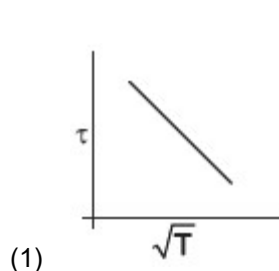
(1) 2×10^{-7} s

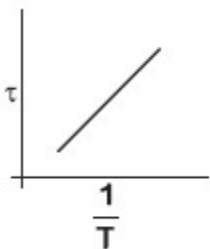
(2) 3×10^{-6} s

(3) 0.5×10^{-8} s

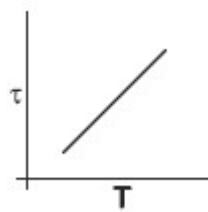
(4) 4×10^{-8} s

7. The plot that depicts the behavior of the mean free time τ (time between two successive collisions) for the molecules of an ideal gas, as a function of temperature (T), qualitatively, is (Graphs are schematic and not drawn to scale)





(3)



(4)

8. An ideal gas in a closed container is slowly heated. As its temperature increases, which of the following statements are true?

- (A) the mean free path of the molecules decreases.
(B) the mean collision time between the molecules decreases.
(C) the mean free path remains unchanged.
(D) the mean collision time remains unchanged

(1) (C) and (D)

(2) (A) and (D)

(3) (B) and (C)

(4) (A) and (B)

9. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: When a rod lying freely is heated, no thermal stress is developed in it.

Reason R: On heating, the length of the rod increases.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is true but R is false
(2) A is false but R is true
(3) Both A and R are true but R is NOT the correct explanation of A
(4) Both A and R are true and R is the correct explanation of A

10. Calculate the value of mean free path (λ) for oxygen molecules at temperature 27°C and pressure 1.01×10^5 Pa. Assume the molecular diameter 0.3 nm and the gas is ideal. ($k = 1.38 \times 10^{-23} \text{ JK}^{-1}$)

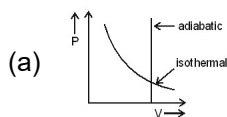
(1) 86 nm

(2) 32 nm

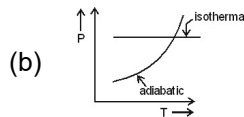
(3) 58 nm

(4) 102 nm

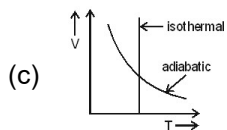
11. Which one is the correct option for the two different thermodynamic processes?



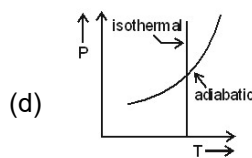
(a)



(b)



(c)



(d)

(1) (c) and (d)

(2) (a) only

(3) (c) and (a)

(4) (b) and (c)

12. For an ideal gas the instantaneous change in pressure 'p' with volume 'v' is given by the equation $\frac{dp}{dv} = -ap$. If $p = p_0$ at $v = 0$ is the given boundary condition, then the maximum temperature one mole of gas can attain is:

(Here R is the gas constant)

- (1) Infinity (2) $\frac{ap_0}{eR}$
(3) 0°C (4) $\frac{p_0}{aeR}$

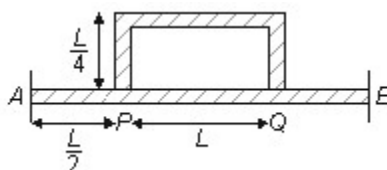
- ** 13.** A Carnot engine having an efficiency of $\frac{1}{10}$ is being used as a refrigerator. If the work done on the refrigerator is 10 J, the amount of heat absorbed from the reservoir at lower temperature is :

- (1) 90 J (2) 1 J
(3) 99 J (4) 100 J

- ** 14.** Two Carnot engines A and B are operated in series. The first one, A, receives heat at $T_1 (=600\text{ K})$ and rejects to a reservoir at temperature T_2 . The second engine B receives heat rejected by the first engine and, in turn, rejects to a heat reservoir at $T_3 (=400\text{ K})$. Calculate the temperature T_2 if the work outputs of the two engines are equal

- (1) 300 K (2) 400 K
(3) 600 K (4) 500 K

15. Temperature difference of 120°C is maintained between two ends of a uniform rod AB of length $2L$. Another bent rod PQ, of same cross-section as AB and length $\frac{3L}{2}$, is connected across AB (see figure). In steady state, temperature difference between P and Q will be close to



- (1) 35°C (2) 45°C
(3) 60°C (4) 75°C

16. A thermally insulated vessel contains 150 g of water at 0°C . Then the air from the vessel is pumped out adiabatically. A fraction of water turns into ice and the rest evaporates at 0°C itself. The mass of evaporated water will be closest to:

(Latent heat of vaporization of water = $2.10 \times 10^6\text{ J kg}^{-1}$ and Latent heat of Fusion of water = $3.36 \times 10^5\text{ J kg}^{-1}$)

- (1) 130 g (2) 150 g
(3) 20 g (4) 35 g



17. The volume V of a given mass of monoatomic gas changes with temperature T according to the relation

$V = KT^{\frac{2}{3}}$. The work done when temperature changes by 90 K will be xR . The value of x is ____.

[R = universal gas constant]



ANSWER KEY UNIT-3

1.	(2)	2.	(2)	3.	(1)	4.	(4)	5.	(1)
6.	(4)	7.	(2)	8.	(3)	9.	(3)	10.	(4)
11.	(1)	12.	(4)	13.	(1)	14.	(4)	15.	(2)
16.	(3)	17.	(60)						

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** - Not the part of both "JEE MAIN" and "JEE ADVANCE" syllabus



Unit-4 (Electromagnetism)

1. The magnetic field of an electromagnetic wave is given by:

$$\vec{B} = 1.6 \times 10^{-6} \cos(2 \times 10^7 z + 6 \times 10^{15} t)(2\hat{i} + \hat{j}) \frac{\text{Wb}}{\text{m}^2}$$

The associated electric field will be:

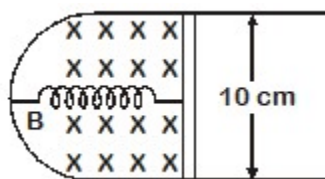
(1) $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z + 6 \times 10^{15} t)(-\hat{i} + 2\hat{j}) \frac{\text{V}}{\text{m}}$

(2) $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z - 6 \times 10^{15} t)(2\hat{i} + \hat{j}) \frac{\text{V}}{\text{m}}$

(3) $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z - 6 \times 10^{15} t)(-2\hat{j} + \hat{i}) \frac{\text{V}}{\text{m}}$

(4) $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z + 6 \times 10^{15} t)(\hat{i} - 2\hat{j}) \frac{\text{V}}{\text{m}}$

2. A thin strip 10 cm long is on a U-shaped wire of negligible resistance and it is connected to a spring of spring constant 0.5 Nm^{-1} (see figure). The assembly is kept in a uniform magnetic field of 0.1 T. If the strip is pulled from its equilibrium position and released, the number of oscillations it performs before its amplitude decreases by a factor of e is N . If the mass of the strip is 50 grams, its resistance 10Ω and air drag negligible, N will be close to



- (1) 1000 (2) 5000
(3) 50000 (4) 10000
3. In a conductor, if the number of conduction electrons per unit volume is $8.5 \times 10^{28} \text{ m}^{-3}$ and mean free time is 25 fs (femto second), its approximate resistivity is

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$(m_e = 9.1 \times 10^{-31} \text{ kg})$

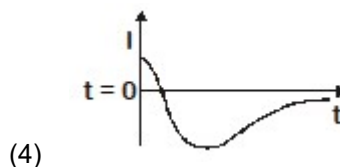
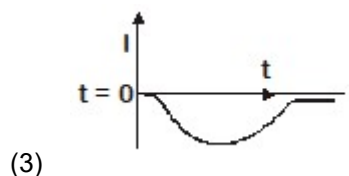
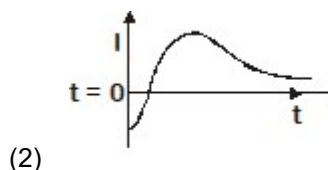
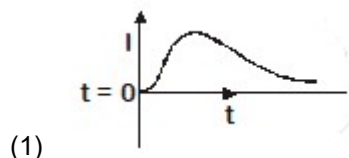
(1) $10^{-5} \Omega\text{m}$

(2) $10^{-6} \Omega\text{m}$

(3) $10^{-7} \Omega\text{m}$

(4) $10^{-8} \Omega\text{m}$

4. A very long solenoid of radius R is carrying current $I(t) = kte^{-\alpha t}$ ($k > 0$), as a function of time ($t \geq 0$). Counter clockwise current is taken to be positive. A circular conducting coil of radius $2R$ is placed in the equatorial plane of the solenoid and concentric with the solenoid. The current induced in the outer coil is correctly depicted, as a function of time, by:



5. The magnetic field of a plane electromagnetic wave is given by :

$$\vec{B} = \vec{B}_0 \hat{i} [\cos(kz - \omega t) + B_1 \hat{j} \cos(kz + \omega t)]$$

where $B_0 = 3 \times 10^{-5} \text{ T}$ and $B_1 = 2 \times 10^{-6} \text{ T}$.

The rms value of the force experienced by a stationary charge $Q = 10^{-4} \text{ C}$ at $z = 0$ is closest to:

(1) 0.6 N

(2) 0.9 N

(3) $3 \times 10^{-2} \text{ N}$

(4) 0.1 N

6. A uniformly charged ring of radius $3a$ and total charge q is placed in xy -plane centred at origin. A point charge q is moving towards the ring along the z -axis and has speed v at $z = 4a$. The minimum value of v such that it crosses the origin is :

(1) $\sqrt{\frac{2}{m} \left(\frac{1}{15} \frac{q^2}{4\pi\epsilon_0 a} \right)^{1/2}}$

(2) $\sqrt{\frac{2}{m} \left(\frac{1}{5} \frac{q^2}{4\pi\epsilon_0 a} \right)^{1/2}}$

(3) $\sqrt{\frac{2}{m} \left(\frac{4}{15} \frac{q^2}{4\pi\epsilon_0 a} \right)^{1/2}}$

(4) $\sqrt{\frac{2}{m} \left(\frac{2}{15} \frac{q^2}{4\pi\epsilon_0 a} \right)^{1/2}}$

7. The electric field of a plane electromagnetic wave is given by

$$\vec{E} = E_0 \hat{i} \cos(kz) \cos(\omega t)$$

The corresponding magnetic field $\vec{B}$ is then given by :

(1) $\vec{B} = \frac{E_0}{c} \hat{j} \sin(kz) \cos(\omega t)$

(2) $\vec{B} = \frac{E_0}{c} \hat{j} \cos(kz) \sin(\omega t)$

(3) $\vec{B} = \frac{E_0}{c} \hat{j} \sin(kz) \sin(\omega t)$

(4) $\vec{B} = \frac{E_0}{c} \hat{k} \sin(kz) \cos(\omega t)$

8. A current of 5 A passes through a copper conductor (resistivity = $1.7 \times 10^{-8} \Omega \text{ m}$) of radius of cross-section 5 mm. Find the mobility of the charges if their drift velocity is $1.1 \times 10^{-3} \text{ m/s}$.

- (1) $1.3 \text{ m}^2/\text{Vs}$ (2) $1.8 \text{ m}^2/\text{Vs}$
(3) $1.5 \text{ m}^2/\text{Vs}$ (4) $1.0 \text{ m}^2/\text{Vs}$

9. Given below in the left column are different modes of communication using the kinds of waves given in the right column.

- | | |
|--------------------------------|-------------------|
| A. Optical Fibre Communication | P. Ultrasound |
| B. Radar | Q. Infrared Light |
| C. Sonar | R. Microwaves |
| D. Mobile Phones | S. Radio Waves |

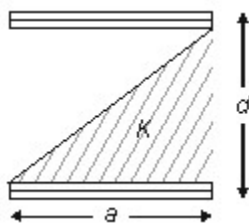
From the options given below, find the most appropriate match between entries in the left and the right column.

- (1) A-Q, B-S, C-P, D-R (2) A-Q, B-S, C-R, D-P
(3) A-S, B-Q, C-R, D-P (4) A-R, B-P, C-S, D-Q

10. A moving coil galvanometer, having a resistance G , produces full scale deflection when a current I_g flows through it. This galvanometer can be converted into (i) an ammeter of range 0 to I_0 ($I_0 > I_g$) by connecting a shunt resistance R_A to it and (ii) into a voltmeter of range 0 to V ($V = GI_0$) by connecting a series resistance R_V to it. Then,

- (1) $R_A R_V = G^2$ and $\frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g} \right)^2$ (2) $R_A R_V = G^2 \left(\frac{I_g}{I_0 - I_g} \right)$ and $\frac{R_A}{R_V} = \left(\frac{I_0 - I_g}{I_g} \right)^2$
(3) $R_A R_V = G^2$ and $\frac{R_A}{R_V} = \frac{I_g}{(I_0 - I_g)}$ (4) $R_A R_V = G^2 \left(\frac{(I_0 - I_g)}{I_g} \right)$ and $\frac{R_A}{R_V} = \left(\frac{I_g}{(I_0 - I_g)} \right)^2$

11. A parallel plate capacitor is made of two square plates of side a , separated by a distance d ($d \ll a$). The lower triangular portion is filled with a dielectric of dielectric constant K , as shown in the figure. Capacitance of this capacitor is



- (1) $\frac{K\epsilon_0 a^2}{d(K-1)} \ln K$ (2) $\frac{K\epsilon_0 a^2}{d} \ln K$
(3) $\frac{K\epsilon_0 a^2}{2d(K+1)}$ (4) $\frac{1}{2} \frac{K\epsilon_0 a^2}{d}$

-
- A diagram showing a vertical line with an upward-pointing arrow. A horizontal arrow labeled 'd' points from the vertical line to a circular arrow, indicating a distance or displacement.

$$(2) \quad F \propto \left(\frac{a}{d}\right)^2$$

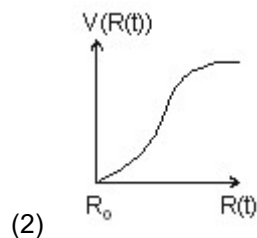
$$(4) \quad F \propto \left(\frac{a^2}{d^3} \right)$$

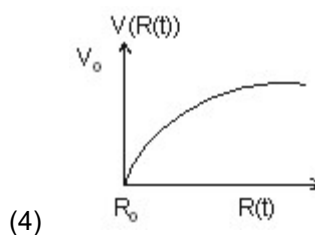
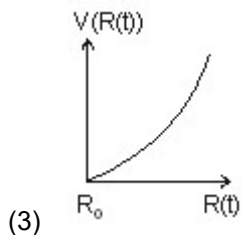
- (4) $3.6 \times 10^{-5} \text{ N}$

- $$(2) \quad \left(\sqrt{n}, \frac{1}{\sqrt{n}} \right)$$

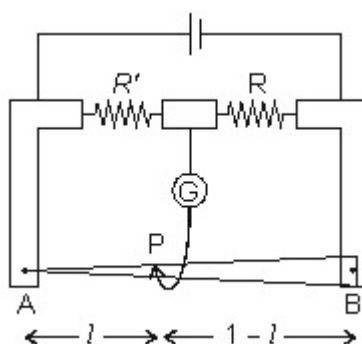
$$(4) \quad (\sqrt{n}, \sqrt{n})$$

- (1) 





16. In a meter bridge, the wire of length 1 m has a non-uniform cross-section such that, the variation $\frac{dR}{dl}$ of its resistance R with length l is $\frac{dR}{dl} \propto \frac{1}{\sqrt{l}}$. Two equal resistances are connected as shown in the figure. The galvanometer has zero deflection when the jockey is at point P . What is the length AP ?



- (1) 0.2 m (2) 0.35 m
(3) 0.25 m (4) 0.3 m
17. A light wave is incident normally on a glass slab of refractive index 1.5. If 4% of light gets reflected and the amplitude of the electric field of the incident light is 30 V/m, then the amplitude of the electric field for the wave propagating in the glass medium will be
- (1) 6 V/m (2) 10 V/m
(3) 24 V/m (4) 30 V/m
18. A long solenoid of radius R carries a time (t) - dependent current $I(t) = I_0 t(1 - t)$. A ring of radius $2R$ is placed coaxially near its middle. During the time interval the induced current (I_R) and the induced EMF (V_R) in the ring change as:
- (1) Direction of I_R remains unchanged and V_R is zero at $t = 0.25$
(2) Direction of I_R remains unchanged and V_R is maximum at $t = 0.5$
(3) At $t = 0.5$ direction of I_R reverses and V_R is zero
(4) At $t = 0.25$ direction of I_R reverses and V_R is maximum
19. The electric field of a plane electromagnetic wave is given by

$$\vec{E} = E_0 \frac{\hat{i} + \hat{j}}{\sqrt{2}} \cos(kz + \omega t)$$

At $t = 0$, a positively charged particle is at the point $(x, y, z) = \left(0, 0, \frac{\pi}{k}\right)$. If its instantaneous velocity at $(t = 0)$

is $v_0 \hat{k}$, the force acting on it due to the wave is:

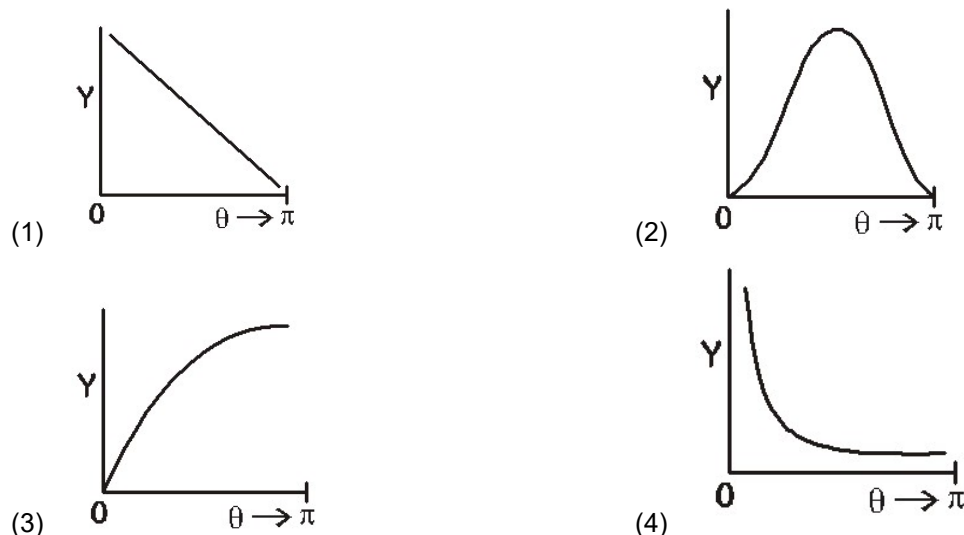
- (1) Antiparallel to $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$ (2) Zero
(3) Parallel to $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$ (4) Parallel to $\hat{k}$

20. The graph which depicts the results of Rutherford gold foil experiment with α -particle is

θ : Scattering angle

Y : Number of scattered α -particles detected

(Plots are schematic and not to scale)



21. The electric fields of two plane electromagnetic plane waves in vacuum are given by

$$\vec{E}_1 = E_0 \hat{j} \cos(\omega t - kx) \text{ and}$$

$$\vec{E}_2 = E_0 \hat{k} \cos(\omega t - ky)$$

At $t = 0$, a particle of charge q is at origin with a velocity $\vec{v} = 0.8 c \hat{j}$ (c is the speed of light in vacuum). The instantaneous force experienced by the particle is:

- (1) $E_0 q (0.4 \hat{i} - 3 \hat{j} + 0.8 \hat{k})$ (2) $E_0 q (-0.8 \hat{i} + \hat{j} + \hat{k})$
(3) $E_0 q (0.8 \hat{i} + \hat{j} + 0.2 \hat{k})$ (4) $E_0 q (0.8 \hat{i} - \hat{j} + 0.4 \hat{k})$

22. A plane electromagnetic wave is propagating along the direction $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$, with its polarization along the direction

$\hat{k}$. The correct form of the magnetic field of the wave would be (here B_0 is an appropriate constant)

$$(1) \quad B_0 \frac{\hat{i} - \hat{j}}{\sqrt{2}} \cos\left(\omega t - k \frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$$

$$(2) \quad B_0 \hat{k} \cos\left(\omega t - k \frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$$

$$(3) \quad B_0 \frac{\hat{i} + \hat{j}}{\sqrt{2}} \cos\left(\omega t - k \frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$$

$$(4) \quad B_0 \frac{\hat{j} - \hat{i}}{\sqrt{2}} \cos\left(\omega t + k \frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$$

23. A plane electromagnetic wave, has frequency of 2.0×10^{10} Hz and its energy density is 1.02×10^{-8} J/m³ in vacuum. The amplitude of the magnetic field of the wave is close to $\left(\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \frac{\text{Nm}^2}{\text{C}^2}\right)$ and speed of light =

$$3 \times 10^8 \text{ ms}^{-1})$$

$$(1) \quad 190 \text{ nT}$$

$$(2) \quad 160 \text{ nT}$$

$$(3) \quad 150 \text{ nT}$$

$$(4) \quad 180 \text{ nT}$$

24. Magnetic materials used for making permanent magnets (P) and magnets in a transformer (T) have different properties of the following, which property best matches for the type of magnet required?

$$(1) \quad T : \text{Large retentivity, small coercivity}$$

$$(2) \quad P : \text{Large retentivity, large coercivity}$$

$$(3) \quad P : \text{Small retentivity, large coercivity}$$

$$(4) \quad T : \text{Large retentivity, large coercivity}$$

25. The fractional change in the magnetic field intensity at a distance 'r' from centre on the axis of current carrying coil of radius 'a' to the magnetic field intensity at the centre of the same coil is :

$$(\text{Take } r < a).$$

$$(1) \quad \frac{3}{2} \frac{r^2}{a^2}$$

$$(2) \quad \frac{2}{3} \frac{a^2}{r^2}$$

$$(3) \quad \frac{2}{3} \frac{r^2}{a^2}$$

$$(4) \quad \frac{3}{2} \frac{a^2}{r^2}$$

26. A paramagnetic sample shows a net magnetisation of 6 A/m when it is placed in an external magnetic field of 0.4 T at a temperature of 4 K. When the sample is placed in an external magnetic field of 0.3 T at a temperature of 24 K, then the magnetisation will be

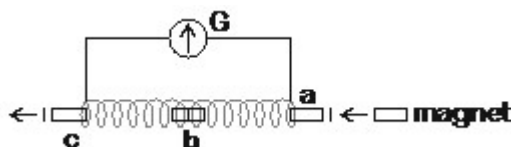
$$(1) \quad 1 \text{ A/m}$$

$$(2) \quad 0.75 \text{ A/m}$$

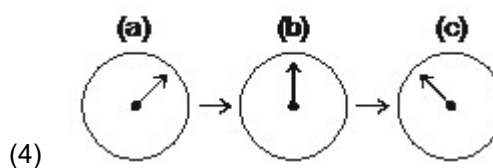
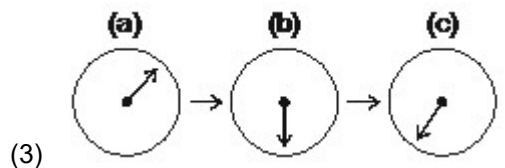
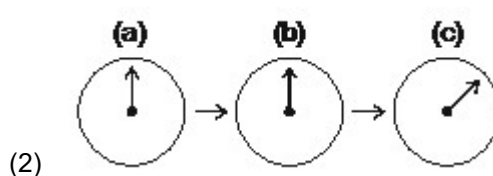
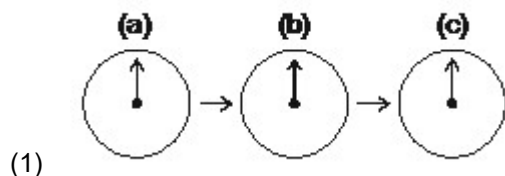
$$(3) \quad 4 \text{ A/m}$$

$$(4) \quad 2.25 \text{ A/m}$$

27. A small bar magnet is moved through a coil at constant speed from one end to the other. Which of the following series of observations will be seen on the galvanometer G attached across the coil?



Three positions shown describe : (a) the magnet's entry (b) magnet is completely inside and (c) magnet's exit.



28. The correct match between the entries in column I and column II are

I

Radiation

- (a) Microwave
- (b) Gamma rays
- (c) A.M. radio waves
- (d) X = rays

II

Wavelength

- (i) 100 m
- (ii) 10^{-15} m
- (iii) 10^{-10} m
- (iv) 10^{-3} m
- (2) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
- (4) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

- (1) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
- (3) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

29. An electron is constrained to move along the y-axis with a speed of $0.1c$ (c is the speed of light) in the presence of electromagnetic wave, whose electric field is $\vec{E} = 30\hat{j} \sin(1.5 \times 10^7 t - 5 \times 10^{-2} x)$ V/m. The maximum magnetic force experienced by the electron will be

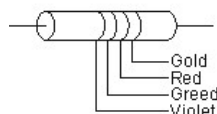
(given $c = 3 \times 10^8 \text{ ms}^{-1}$ and electron charge = $1.6 \times 10^{-19} \text{ C}$)

- (1) $4.8 \times 10^{-19} \text{ N}$
- (2) $2.4 \times 10^{-18} \text{ N}$
- (3) $3.2 \times 10^{-18} \text{ N}$
- (4) $1.6 \times 10^{-19} \text{ N}$

30. A circuit to verify Ohm's law uses ammeter and voltmeter in series or parallel connected correctly to the resistor. In the circuit

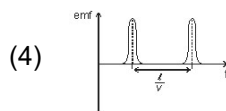
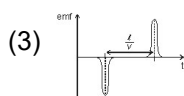
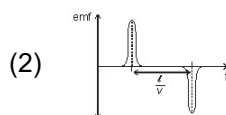
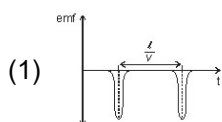
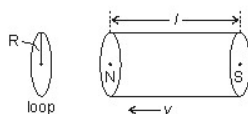
- (1) Ammeter is always connected in series and voltmeter in parallel
- (2) Both ammeter and voltmeter must be connected in parallel
- (3) Ammeter is always used in parallel and voltmeter is series
- (4) Both ammeter and voltmeter must be connected in series

*** 31. The colour coding on a carbon resistor is shown in the given figure. The resistance value of the given resistor is:



- (1) $(7500 \pm 750) \Omega$
- (2) $(5700 \pm 375) \Omega$
- (3) $(5700 \pm 285) \Omega$
- (4) $(7500 \pm 375) \Omega$

32. A bar magnet is passing through a conducting loop of radius R with velocity v . The radius of the bar magnet is such that it just passes through the loop. The induced e.m.f. in the loop can be represented by the approximate curve :



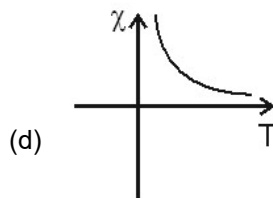
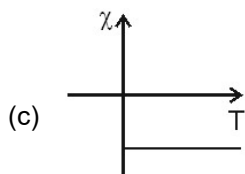
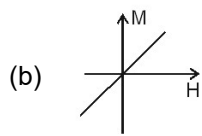
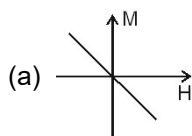
33. Choose the incorrect statement:

- The electric lines of force entering into a Gaussian surface provide negative flux.
- A charge 'q' is placed at the centre of a cube. The flux through all the faces will be the same.
- In a uniform electric field net flux through a closed Gaussian surface containing no net charge, is zero.
- When electric field is parallel to a Gaussian surface, it provides a finite non-zero flux.

Choose the most appropriate answer from the options given below:

- (d) Only
- (c) and (d) Only
- (a) and (c) Only
- (b) and (d) Only

34. Following plots show Magnetization (M) vs Magnetising field (H) and Magnetic susceptibility χ vs Temperature (T) graph:



Which of the following combination will be represented by a diamagnetic material?

- (a), (d)
- (b), (c)
- (b), (d)
- (a), (c)

35. Electric field of a plane electromagnetic wave propagating through a non-magnetic medium is given by $E = 20\cos(2 \times 10^{10} t - 200x)$ V/m. The dielectric constant of the medium is equal to :

(Take $\mu_r = 1$)

(1) $\frac{1}{3}$

(2) 9

(3) 3

(4) 2

36. A soft ferromagnetic material is placed in an external magnetic field. The magnetic domains:

- (1) decrease in size and changes orientation.
- (2) may increase or decrease in size and change its orientation.
- (3) increase in size but no change in orientation.
- (4) have no relation with external magnetic field

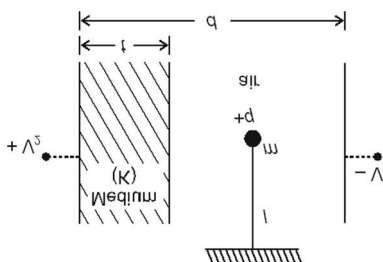
37. In a ferromagnetic material, below the curie temperature, a domain is defined as :

- (1) a macroscopic region with randomly oriented magnetic dipoles.
- (2) a macroscopic region with consecutive magnetic dipoles oriented in opposite direction.
- (3) a macroscopic region with saturation magnetization.
- (4) a macroscopic region with zero magnetization.

38. Choose the correct option

- (1) True dip is always equal to apparent dip.
- (2) True dip is not mathematically related to apparent dip.
- (3) True dip is less than the apparent dip.
- (4) True dip is always greater than the apparent dip.

39. A simple pendulum of mass ' m ', length ' l ' and charge '+ q ' suspended in the electric field produced by two conducting parallel plates as shown. The value of deflection of pendulum in equilibrium position will be



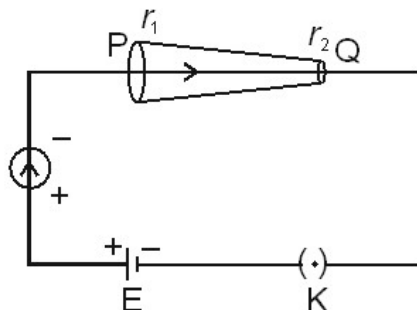
(1) $\tan^{-1} \left[\frac{q}{mg} \times \frac{C_2(V_2 - V_1)}{(C_1 + C_2)(d - t)} \right]$

(2) $\tan^{-1} \left[\frac{q}{mg} \times \frac{C_1(V_2 - V_1)}{(C_1 + C_2)(d - t)} \right]$

(3) $\tan^{-1} \left[\frac{q}{mg} \times \frac{C_1(V_1 + V_2)}{(C_1 + C_2)(d - t)} \right]$

(4) $\tan^{-1} \left[\frac{q}{mg} \times \frac{C_2(V_1 + V_2)}{(C_1 + C_2)(d - t)} \right]$

40. In the given figure, a battery of emf E is connected across a conductor PQ of length ' l ' and different area of cross-sections having radii r_1 and r_2 ($r_2 < r_1$).



Choose the correct option as one moves from P to Q

- (1) All of these (2) Electron current decreases
 (3) Electric field decreases (4) Drift velocity of electron increases

41. Which of the following statements are correct?

- (A) Electric monopoles do not exist whereas magnetic monopoles exist.
 (B) Magnetic field lines due to a solenoid at its ends and outside cannot be completely straight and confined.
 (C) Magnetic field lines are completely confined within a toroid.
 (D) Magnetic field lines inside a bar magnet are not parallel.
 (E) $x = -1$ is the condition for a perfect diamagnetic material, where x is its magnetic susceptibility.

Choose the correct answer from the options given below:

- (1) (C) and (E) only (2) (B) and (C) only
 (3) (B) and (D) only (4) (A) and (B) only

42. The peak electric field produced by the radiation coming from the 8 W bulb at a distance of 10 m is $\frac{x}{10} \sqrt{\frac{\mu_0 c}{\pi}} \frac{V}{m}$.

The efficiency of the bulb is 10% and it is a point source. The value of x is _____.

ANSWER KEY

UNIT-4



1.	(1)	2.	(2)	3.	(4)	4.	(2)	5.	(1)
6.	(4)	7.	(3)	8.	(4)	9.	(1)	10.	(1)
11.	(1)	12.	(2)	13.	(3)	14.	(2)	15.	(4)
16.	(3)	17.	(3)	18.	(3)	19.	(1)	20.	(4)
21.	(3)	22.	(1)	23.	(2)	24.	(2)	25.	(1)
26.	(2)	27.	(4)	28.	(2)	29.	(1)	30.	(1)
31.	(4)	32.	(3)	33.	(1)	34.	(4)	35.	(2)
36.	(2)	37.	(3)	38.	(3)	39.	(4)	40.	(4)
41.	(3)	42.	(02)						



Unit-5 (Optics and Modern Physics)

- ** 1.** The physical sizes of the transmitter and receiver antenna in a communication system are:
- (1) Inversely proportional to modulation frequency
 - (2) Inversely proportional to carrier frequency
 - (3) Proportional to carrier frequency
 - (4) Independent of both carrier and modulation frequency
- 2.** Diameter of the objective lens of a telescope is 250 cm. For light of wavelength 600 nm. coming from a distant object, the limit of resolution of the telescope is close to :
- (1) 1.5×10^{-7} rad
 - (2) 3.0×10^{-7} rad
 - (3) 2.0×10^{-7} rad
 - (4) 4.5×10^{-7} rad
- ** 3.** An NPN transistor is used in common emitter configuration as an amplifier with $1 \text{ k}\Omega$ load resistance. Signal voltage of 10 mV is applied across the base-emitter. This produces a 3 mA change in the collector current and $15 \mu\text{A}$ change in the base current of the amplifier. The input resistance and voltage gain are :
- (1) $0.33 \text{ k}\Omega$, 1.5
 - (2) $0.33 \text{ k}\Omega$, 300
 - (3) $0.67 \text{ k}\Omega$, 200
 - (4) $0.67 \text{ k}\Omega$, 300
- ** 4.** An npn transistor operates as a common emitter amplifier, with a power gain of 60 dB. The input circuit resistance is 100Ω and the output load resistance is $10 \text{ k}\Omega$. The common emitter current gain b is :
- (1) 10^4
 - (2) 6×10^2
 - (3) 10^2
 - (4) 60
- 5.** A system of three polarizers P_1 , P_2 , P_3 is set up such that the pass axis of P_3 is crossed with respect to that of P_1 . The pass axis of P_2 is inclined at 60° to the pass axis of P_3 . When a beam of unpolarized light of intensity I_0 is incident on P_1 , the intensity of light transmitted by the three polarizers is I . The ratio (I_0/I) equals (nearly) :
- (1) 1.80
 - (2) 5.33
 - (3) 10.67
 - (4) 16.00
- 6.** The truth table for the circuit given in the fig. is:



(1)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

(2)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

(3)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	1

(4)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

- *** 7. In a communication system operating at wavelength 800 nm, only one percent of source frequency is available as signal bandwidth. The number of channels accommodated for transmitting TV signals of band width 6 MHz are (take velocity of light $c = 3 \times 10^8 \text{ m/s}$, $h = 6.6 \times 10^{-34} \text{ J-s}$)

- (1) 3.75×10^6 (2) 3.86×10^6
(3) 6.25×10^5 (4) 4.87×10^5

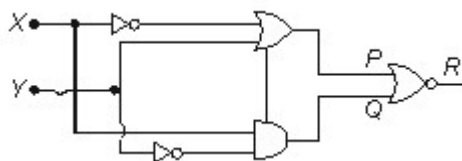
8. Mobility of electrons in a semiconductor is defined as the ratio of their drift velocity to the applied electric field. If, for an n -type semiconductor, the density of electrons is 10^{19} m^{-3} and their mobility is $1.6 \text{ m}^2/(\text{V.s})$ then the resistivity of the semiconductor (since it is an n -type semiconductor contribution of holes is ignored) is close

- (1) $2 \Omega\text{m}$ (2) $0.2 \Omega\text{m}$
(3) $0.4 \Omega\text{m}$ (4) $4 \Omega\text{m}$

- *** 9. The modulation frequency of an AM radio station is 250 kHz, which is 10% of the carrier wave. If another AM station approaches you for license what broadcast frequency will you allot?

- (1) 2750 kHz (2) 2900 kHz
(3) 2000 kHz (4) 2250 kHz

10. To get output '1' at R, for the given logic gate circuit the input values must be

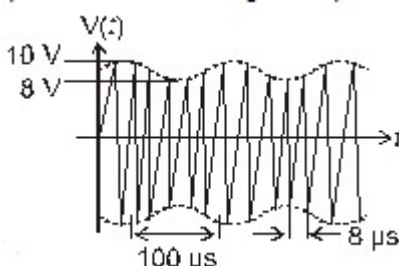


- (1) $X = 1, Y = 1$ (2) $X = 0, Y = 0$
(3) $X = 1, Y = 0$ (4) $X = 0, Y = 1$

11. In an electron microscope, the resolution that can be achieved is of the order of the wavelength of electrons used. To resolve a width of $7.5 \times 10^{-12} \text{ m}$, the minimum electron energy required is close to

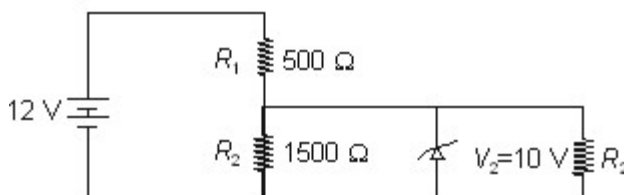
- (1) 100 keV (2) 1 keV
(3) 500 keV (4) 25 keV

- ** 12. An amplitude modulated signal is plotted below:

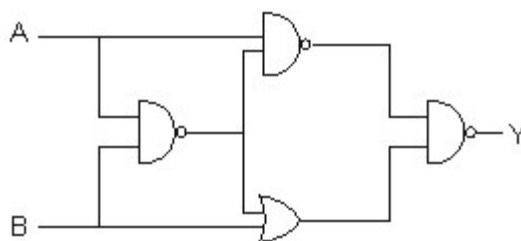


Which one of the following best describes the above signal?

- (1) $(9 + \sin(2\pi \times 10^4 t))\sin(2.5\pi \times 10^5 t)$ V (2) $(9 + \sin(4\pi \times 10^4 t))\sin(5\pi \times 10^5 t)$ V
(3) $(1 + 9\sin(2\pi \times 10^4 t))\sin(2.5\pi \times 10^5 t)$ V (4) $(9 + \sin(2.5\pi \times 10^5 t))\sin(2\pi \times 10^4 t)$ V
13. In the given circuit the current through Zener Diode is close to



- (1) 6.7 mA (2) 0.0 mA
(3) 4.0 mA (4) 6.0 mA
14. The output of the given logic circuit is

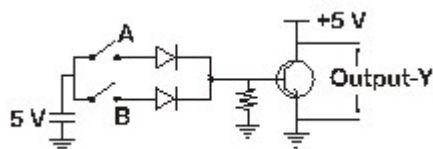


- (1) $A\bar{B} + \bar{A}B$ (2) $A\bar{B}$
(3) $AB + \bar{A}\bar{B}$ (4) $\bar{A}\bar{B}$
15. Visible light of wavelength 6000×10^{-8} cm falls normally on a single slit and produces a diffraction pattern. It is found that the second diffraction minimum is at 60° from the central maximum. If the first minimum is produced at θ_1 , then θ_1 is close to
- (1) 25° (2) 30°
(3) 20° (4) 45°
16. A polarizer-analyser set is adjusted such that the intensity of light coming out of the analyser is just 10% of the original intensity. Assuming the polarizer - analyser set does not absorb any light, the angle by which the analyser need to be rotated further to reduced the output intensity to be zero, is
- (1) 71.6° (2) 45°

(3) 90°

(4) 18.4°

**** 17.** Boolean relation at the output stage-Y for the following circuit is :



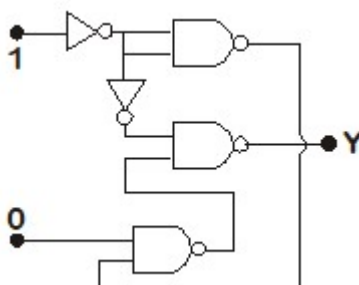
(1) $\bar{A} \cdot \bar{B}$

(2) $A \cdot B$

(3) $A + B$

(4) $\bar{A} + \bar{B}$

18. In the given circuit, value of Y is



(1) Toggles between 0 and 1

(2) 0

(3) 1

(4) Will not execute

**** 19.** The aperture diameter of a telescope is 5 m. The separation between the moon and the earth is 4×10^5 km. With light of wavelength of $5500 \text{ \AA}$, the minimum separation between objects on the surface of moon, so that they are just resolved, is close to:

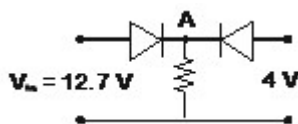
(1) 20 m

(2) 200 m

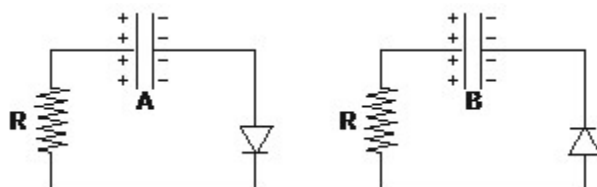
(3) 600 m

(4) 60 m

20. Both the diodes used in the circuit shown are assumed to be ideal and have negligible resistance when these are forward biased. Built in potential in each diode is 0.7 V. For the input voltages shown in the figure, the voltage (in Volts) at point A is _____



21. Two identical capacitors A and B, charged to the same potential 5V are connected in two different circuits as shown below at time $t = 0$. If the charge on capacitors A and B at time $t = CR$ is Q_A and Q_B respectively, then (Here e is the base of natural logarithm)



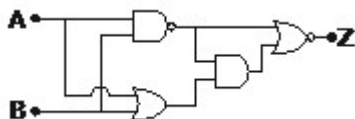
$$(1) \quad Q_A = \frac{CV}{2}, Q_B = \frac{VC}{e}$$

$$(2) \quad Q_A = VC, Q_B = CV$$

$$(3) \quad Q_A = \frac{CV}{e}, Q_B = \frac{VC}{2}$$

$$(4) \quad Q_A = VC, Q_B = \frac{VC}{e}$$

22. In the following digital circuit, what will be the output at 'Z', when the input (A, B) are (1, 0), (0, 0), (1, 1), (0, 1)



$$(1) \quad 1, 1, 0, 1$$

$$(2) \quad 0, 0, 1, 0$$

$$(3) \quad 1, 0, 1, 1$$

$$(4) \quad 0, 1, 0, 0$$

23. The least count of the main scale of a vernier callipers is 1 mm. Its vernier scale is divided into 10 divisions and coincide with 9 divisions of the main scale. When jaws are touching each other, the 7th division of vernier scale coincides with a division of main scale and the zero of vernier scale is lying right side of the zero of main scale. When this vernier is used to measure length of a cylinder the zero of the vernier scale between 3.1 cm and 3.2 cm and 4th VSD coincides with a main scale division. The length of the cylinder is (VSD is vernier scale division)

$$(1) \quad 3.21 \text{ cm}$$

$$(2) \quad 2.99 \text{ cm}$$

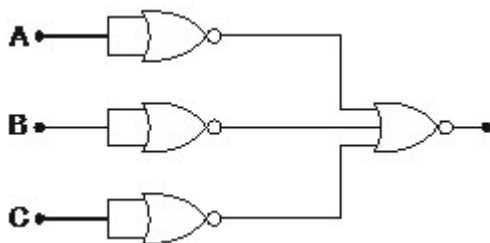
$$(3) \quad 3.07 \text{ cm}$$

$$(4) \quad 3.2 \text{ cm}$$

- * 24. Which of the following will NOT be observed when a multimeter (operating in resistance measuring mode) probes connected across a component, are just reversed?

- (1) Multimeter shows NO deflection in both cases i.e. before and after reversing the probes if the chosen component is capacitor.
- (2) Multimeter shows NO deflection in both cases i.e. before and after reversing the probes if the chosen component is metal wire.
- (3) Multimeter shows a deflection, accompanied by a splash of light out of connected component in one direction and NO deflection on reversing the probes if the chosen component is LED.
- (4) Multimeter shows an equal deflection in both cases i.e. before and after reversing the probes if the chosen component is resistor.

25. Identify the operation performed by the circuit given below.



$$(1) \quad \text{AND}$$

$$(2) \quad \text{NOT}$$

(3) NAND

(4) OR

26. Your friend is having eye sight problem. She is not able to see clearly a distant uniform window mesh and it appears to her as non-uniform and distorted. The doctor diagnosed the problem as :

(1) Myopia and hypermetropia

(2) Presbyopia with Astigmatism

(3) Myopia with Astigmatism

(4) Astigmatism

27. A beam of plane polarised light of large cross-sectional area and uniform intensity of 3.3 Wm^{-2} falls normally on a polariser (cross sectional area $3 \times 10^{-4} \text{ m}^2$) which rotates about its axis with an angular speed of 31.4 rad/s . The energy of light passing through the polariser per revolution, is close to

(1) $1.0 \times 10^{-5} \text{ J}$

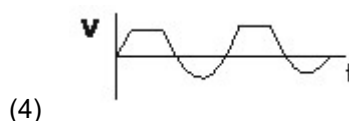
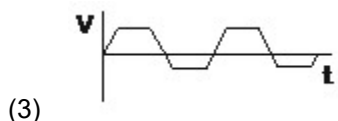
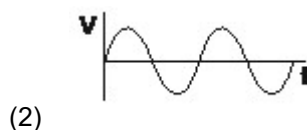
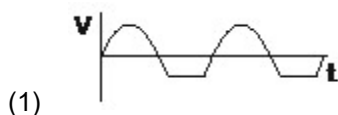
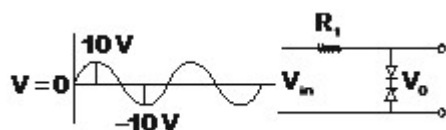
(2) $1.0 \times 10^{-4} \text{ J}$

(3) $5.0 \times 10^{-4} \text{ J}$

(4) $1.5 \times 10^{-4} \text{ J}$

28. Take the breakdown voltage of the zener diode used in the given circuit as 6 V . For the input voltage shown in figure below, the time variation of the output voltage is

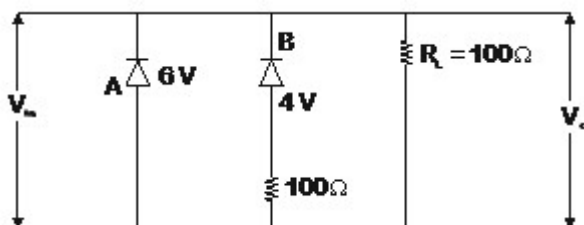
(Graphs drawn are schematic and not to scale)

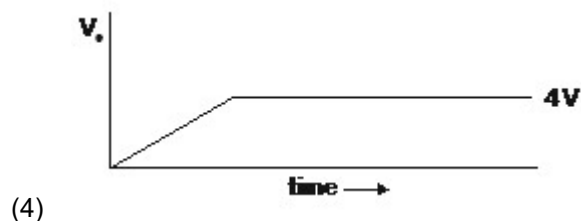
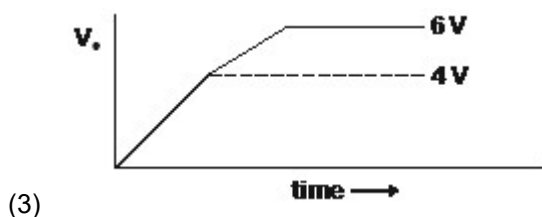
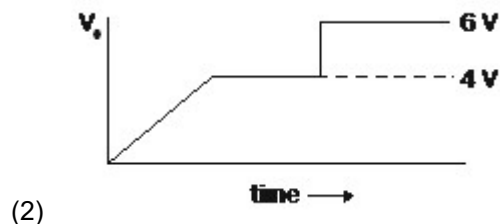
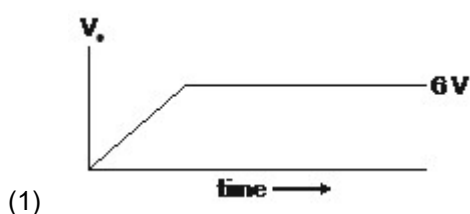


29. Two Zener diodes (A and B) having breakdown voltages of 6 V and 4 V respectively, are connected as shown in the circuit below. The output voltage V_0 variation with input voltage linearly increasing with time, is given by :

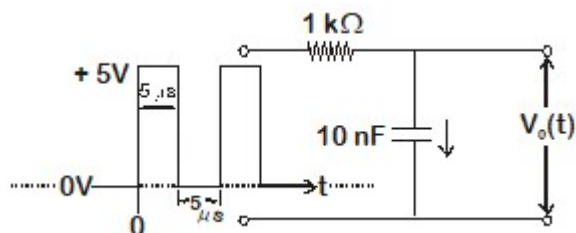
($V_{\text{input}} = 0 \text{ V}$ at $t = 0$)

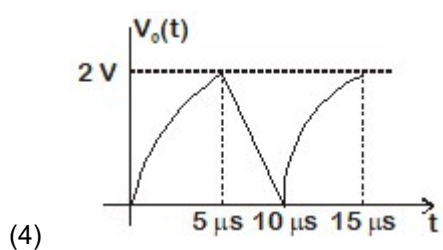
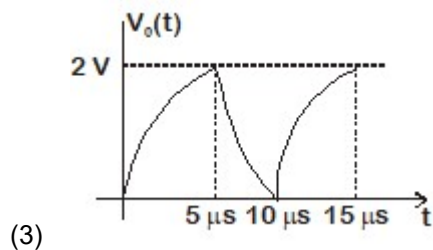
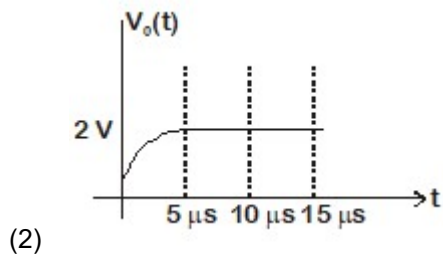
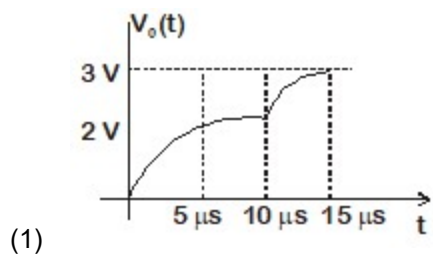
(figures are qualitative)



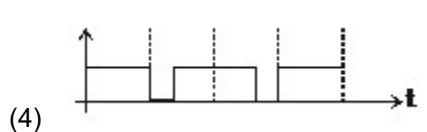
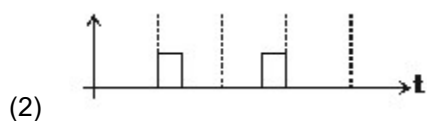
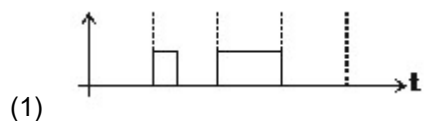
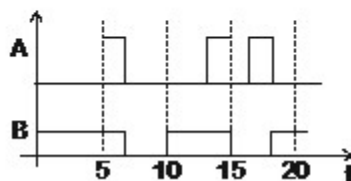


30. With increasing biasing voltage of a photodiode, the photocurrent magnitude
- (1) Increases initially and after attaining certain value, it decreases
 - (2) Increases linearly
 - (3) Increases initially and saturates finally
 - (4) Remains constant
31. Given the masses of various atomic particles $m_p = 1.0072 \text{ u}$, $m_n = 1.0087 \text{ u}$, $m_e = 0.000548 \text{ u}$, where $p \equiv$ proton, $n \equiv$ neutron, $e \equiv$ electron, $\bar{\nu} \equiv$ antineutrino and $d \equiv$ deuteron. Which of the following process is allowed by momentum and energy conservation?
- (1) $n + n \rightarrow \text{deuterium atom}$
(electron bound to the nucleus)
 - (2) $n + p \rightarrow d + \gamma$
 - (3) $p \rightarrow n + e^+ + \bar{\nu}$
 - (4) $e^+ + e^- \rightarrow \gamma$
32. For the given input voltage waveform $V_{in}(t)$, the output voltage waveform $V_o(t)$, across the capacitor is correctly depicted by

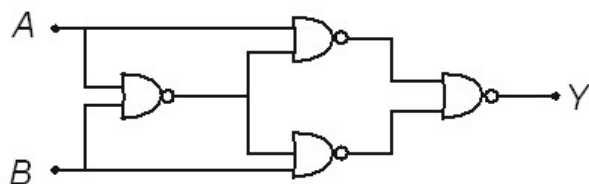




33. Identify the correct output signal Y in the given combination of gates (as shown) for the given inputs A and B.



34. Four NOR gates are connected as shown in figure. The truth table for the given figure is:



AUROS

A C A D E M Y

	A	B	Y
	0	0	1
(1)	0	1	0
	1	0	0
	1	1	1

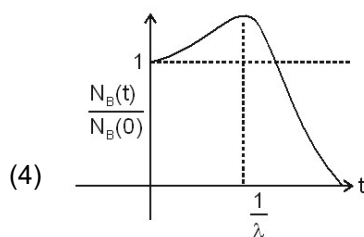
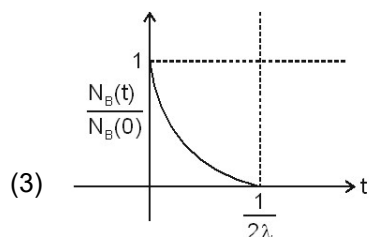
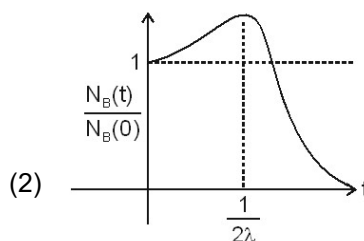
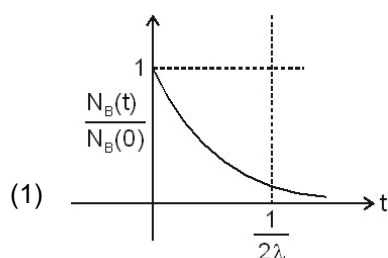
	A	B	Y
	0	0	0
(3)	0	1	1
	1	0	1
	1	1	0

	A	B	Y
	0	0	1
(2)	0	1	0
	1	0	1
	1	1	0

	A	B	Y
	0	0	0
(4)	0	1	1
	1	0	0
	1	1	1

- * 35. At time $t = 0$, a material is composed of two radioactive atoms A and B, where $N_A(0) = 2N_B(0)$. The decay constant of both kind of radioactive atoms is λ . However, A disintegrates to B and B disintegrates to C. Which of the following figures represents the evolution of $N_B(t)/N_B(0)$ with respect to time t ?

$$\begin{aligned} [N_A(0) &= \text{No. of A atoms at } t = 0] \\ [N_B(0) &= \text{No. of B atoms at } t = 0] \end{aligned}$$



36. **Statement I:**

By doping silicon semiconductor with pentavalent material, the electrons density increases.

Statement II:

The n-type semiconductor has net negative charge.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Both statement I and statement II are true
- (2) Both statement I and statement II are false
- (3) Statement I is true but statement II is false
- (4) Statement I is false but statement II is true

**** 37.** For a transistor α and β are given as $\alpha = \frac{I_C}{I_E}$ and $\beta = \frac{I_C}{I_B}$. Then the correct relation between α and β will be:

(1) $\alpha = \frac{1-\beta}{\beta}$

(2) $\beta = \frac{\alpha}{1-\alpha}$

(3) $\alpha\beta = 1$

(4) $\alpha = \frac{\beta}{1-\beta}$

**** 38.** For a transistor in CE mode to be used as an amplifier, it must be operated in:

(1) Cut-off region only

(2) Saturation region only

(3) Both cut-off and saturation

(4) The active region only

39. In a photoelectric experiment, increasing the intensity of incident light :

(1) Increases the frequency of photons incident and the K.E. of the ejected electrons remains unchanged.

(2) Increases the frequency of photons incident and increases the K.E. of the ejected electrons.

(3) Increases the number of photons incident and also increases the K.E. of the ejected electrons.

(4) Increases the number of photons incident and the K.E. of the ejected electrons remains unchanged.

40. Statement I :

To get a steady dc output from the pulsating voltage received from a full wave rectifier we can connect a capacitor across the output parallel to the load R_L .

Statement II :

To get a steady dc output from the pulsating voltage received from a full wave rectifier we can connect an inductor in series with R_L .

In the light of the above statements, choose the **most appropriate** answer from the options given below:

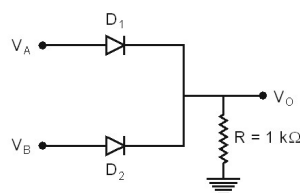
(1) Both statement I and statement II are true

(2) Statement I is false but statement II is true

(3) Both statement I and statement II are false

(4) Statement I is true but statement II is false

**** 41.** If V_A and V_B are the input voltages (either 5 V or 0 V) and V_O is the output voltage then the two gates represented in the following circuits (A) and (B) are :



(A)



- (4)
- 
- A graph showing the number of atoms (y-axis) versus time (x-axis). The curve starts at a high value on the y-axis and decreases exponentially towards the x-axis, representing a decay process.

-
- The diagram shows a 2-to-1 multiplexer implemented with an AND-OR inverter structure. It has two inputs, A and B. Input A is connected to both an AND gate and an OR gate. Input B is connected to the AND gate and the OR gate. The output of the AND gate is labeled P, and the output of the OR gate is labeled Q. The final output Y is the result of an OR operation between P and Q, indicated by a small circle at the output of the OR gate.

- ** - Not the part of both "JEE MAIN" and "JEE ADVANCE" syllabus

Physical Quantity	Least count of the Equipment used for measurement	Observed Value
Mass (M)	1 g	2 kg
Length of bar (L)	1 mm	1 m
Breadth of bar (b)	0.1 mm	4 cm
Thickness of bar (d)	0.01 mm	0.4 cm
Depression (δ)	0.01 mm	5 mm

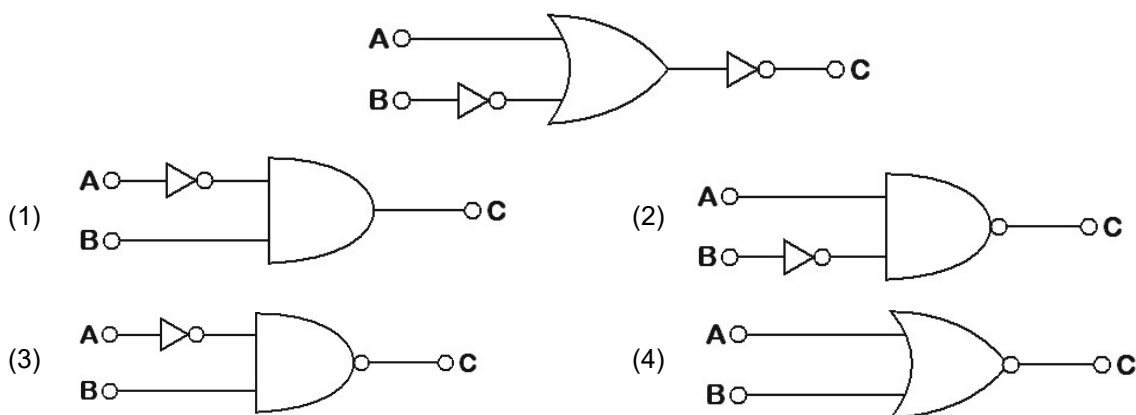
Then the fractional error in the measurement of Y is :

- (1) 0.155 (2) 0.0083
(3) 0.0155 (4) 0.083

45. Zener breakdown occurs in a p-n junction having p and n both

- (1) Lightly doped and have narrow depletion layer
(2) Heavily doped and have narrow depletion layer
(3) Heavily doped and have wide depletion layer
(4) Lightly doped and have wide depletion layer

46. The logic circuit shown above is equivalent to



47. Given below are two statements :

Statement I : PN junction diodes can be used to function as transistor, simply by connecting two diodes, back to back, which acts as the base terminal.

Statement II : In the study of transistor, the amplification factor β indicates ratio of the collector current to the base current.

In the light of the above statements, choose the correct answer from the options given below.

- (1) Both Statement I and Statement II are false (2) Statement I is true but Statement II is false
(3) Statement I is false but Statement II is true (4) Both statement I and Statement II are true

48. Match List-I with List-II.

List-I

- (a) Source of

List-II

- (i) Radioactive decay



- | | |
|----------------------------------|---------------------------------------|
| microwave frequency | of nucleus |
| (b) Source of infrared frequency | (ii) Magnetron |
| (c) Source of Gamma Rays | (iii) Inner shell electrons |
| (d) Source of X-rays | (iv) Vibration of atoms and molecules |
| | (v) LASER |
| | (vi) RC circuit |

Choose the correct answer from the options given below :

- | | |
|---|--|
| (1) (a)-(ii), (b)-(iv), (c)-(vi), (d)-(iii) | (2) (a)-(vi), (b)-(v), (c)-(i), (d)-(iv) |
| (3) (a)-(vi), (b)-(iv), (c)-(i), (d)-(v) | (4) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii) |

49. Match List I with List II.

- | List-I | List-II |
|-----------------|--|
| (a) Rectifier | (i) Used either for stepping up or stepping down the a.c. voltage |
| (b) Stabilizer | (ii) Used to convert a.c. voltage into d.c. voltage |
| (c) Transformer | (iii) Used to remove any ripple in the rectified output voltage |
| (d) Filter | (iv) Used for constant output voltage even when the input voltage or load current change |

Choose the correct answer from the options given below :

- | | |
|--|--|
| (1) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv) | (2) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii) |
| (3) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii) | (4) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii) |

50. Consider the diffraction pattern obtained from the sunlight incident on a pinhole of diameter $0.1 \mu\text{m}$. If the diameter of the pinhole is slightly increased, it will affect the diffraction pattern such that

- | | |
|---|---|
| (1) Its size decreases, and intensity decreases | (2) Its size increases, and intensity increases |
| (3) Its size increases, but intensity decreases | (4) Its size decreases, but intensity increases |

51. For extrinsic semiconductors; when doping level is increased;

- (1) Fermi-level of p-type semiconductors will go downward and Fermi-level of n-type semiconductor will go upward.
- (2) Fermi-level of p and n-type semiconductors will not be affected.
- (3) Fermi-level of both p-type and n-type semiconductors will go upward for $T > T_F$ K and downward for $T < T_F$ K, where T_F is Fermi temperature.
- (4) Fermi-level of p-type semiconductor will go upward and Fermi-level of n-type semiconductors will go downward.

52. Given below are two statements :

Statement-I : A speech signal of 2 kHz is used to modulate a carrier signal of 1 MHz. The band width requirement for the signal is 4 kHz.

Statement II-: The side band frequencies are 1002 kHz and 998 kHz.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both Statement I and Statement II are true (2) Statement I is false but Statement II is true
(3) Statement I is true but Statement II is false (4) Both Statement I and Statement II are false

53. Given below are two statements: one is labeled as Assertion A and the other is labeled as Reason R.

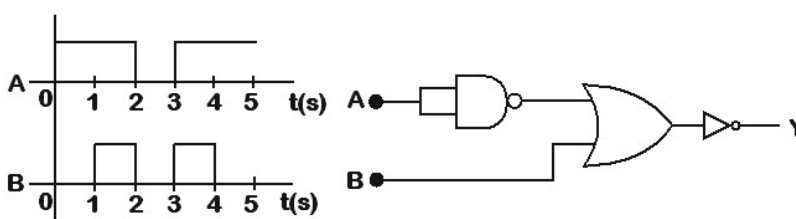
Assertion A : For a simple microscope, the angular size of the object equals the angular size of the image.

Reason R : Magnification is achieved as the small object can be kept much closer to the eye than 25 cm and hence it subtends a large angle.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) A is false but R is true
(2) A is true but R is false
(3) Both A and R are true but R is NOT the correct explanation of A
(4) Both A and R are true and R is the correct explanation of A

54. Draw the output signal Y in the given combination of gates.



- (1) (2)
(3) (4)

55. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : An electron microscope can achieve better resolving power than an optical microscope.

Reason R : The de Broglie's wavelength of the electrons emitted from an electron gun is much less than wavelength of visible light.

In the light of the above statements, choose the correct answer from the options given below :

- (1) A is false but R is true.
(2) Both A and R are true and R is the correct explanation of A.
(3) Both A and R are true but R is NOT the correct explanation of A.
(4) A is true but R is false.

56. A nucleus of mass M emits γ -ray photon of frequency ' ν '. The loss of internal energy by the nucleus is:

[Take 'c' as the speed of electromagnetic wave]

(1) $h\nu \left[1 - \frac{h\nu}{2Mc^2} \right]$ (2) $h\nu$

(3) $h\nu \left[1 + \frac{h\nu}{2Mc^2} \right]$ (4) 0

57. In the Young's double slit experiment, the distance between the slits varies in time as $d(t) = d_0 + a_0 \sin \omega t$; where d_0 , ω and a_0 are constants. The difference between the largest fringe width and the smallest fringe width obtained over time is given as:

(1) $\frac{2\lambda D(d_0)}{(d_0^2 - a_0^2)}$ (2) $\frac{\lambda D}{d_0 + a_0}$

(3) $\frac{2\lambda D a_0}{(d_0^2 - a_0^2)}$ (4) $\frac{\lambda D}{d_0^2 a_0}$

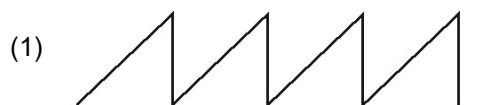
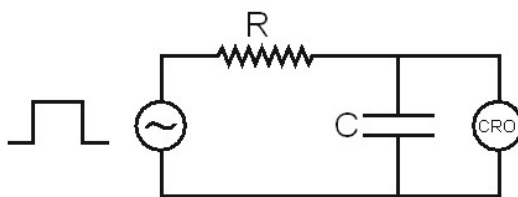
58. Consider the following statements

- A. Atoms of each element emit characteristics spectrum.
- B. According to Bohr's Postulate, an electron in a hydrogen atom, revolves in a certain stationary orbit.
- C. The density of nuclear matter depends on the size of the nucleus.
- D. A free neutron is stable but a free proton decay is possible.
- E. Radioactivity is an indication of the instability of nuclei.

Choose the correct answer from the options given below

- (1) B and D only (2) A, B and E only
(3) A, B, C, D and E (4) A, C and E only

59. An RC circuit as shown in the figure is driven by a AC source generating a square wave. The output wave pattern monitored by CRO would look close to :



60. The speed of electrons in a scanning electron microscope is $1 \times 10^7 \text{ ms}^{-1}$. If the protons having the same speed are used instead of electrons, then the resolving power of scanning proton microscope will be changed by a factor of

(1) $\frac{1}{\sqrt{1837}}$

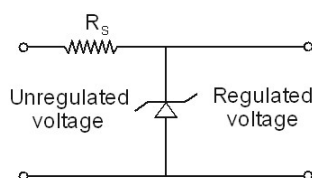
(2) $\frac{1}{1837}$

(3) $\sqrt{1837}$

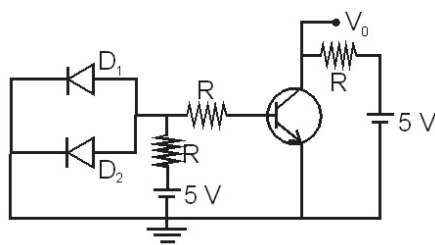
(4) 1837

61. Orange light of wavelength $6000 \times 10^{-10} \text{ m}$ illuminates a single slit of width $0.6 \times 10^{-4} \text{ m}$. The maximum possible number of diffraction minima produced on both sides of the central maximum is _____.

62. A zener diode of power rating 2 W is to be used as a voltage regulator. If the zener diode has a breakdown of 10 V and it has to regulate voltage fluctuated between 6 V and 14 V, the value of R_S for safe operation should be _____ Ω .

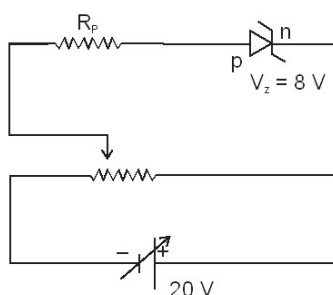


63. A circuit is arranged as shown in figure. The output voltage V_0 is equal to ____ V.

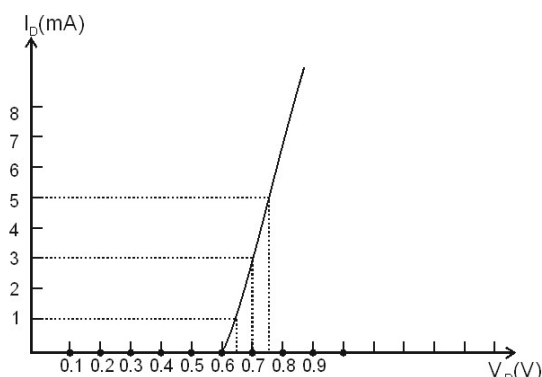


64. A bandwidth of 6 MHz is available for A.M. transmission. If the maximum audio signal frequency used for modulating the carrier wave is not to exceed 6 kHz. The number of stations that can be broadcasted within this band simultaneously without interfering with each other will be _____.

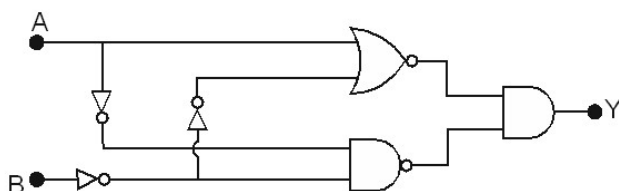
65. A zener diode having zener voltage 8 V and power dissipation rating of 0.5 W is connected across a potential divider arranged with maximum potential drop across zener diode is as shown in the diagram. The value of protective resistance R_P is _____ Ω .



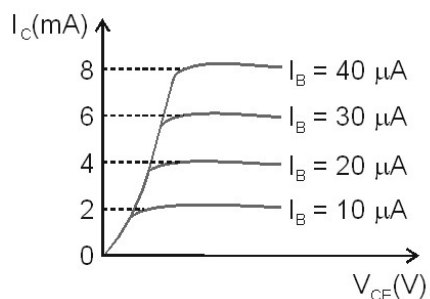
66. For the forward biased diode characteristics shown in the figure, the dynamic resistance at $I_D = 3 \text{ mA}$ will be _____ Ω .



67. In a semiconductor, the number density of intrinsic charge carriers at 27°C is $1.5 \times 10^{16}/\text{m}^3$. If the semiconductor is doped with impurity atom, the hole density increases to $4.5 \times 10^{22}/\text{m}^3$. The electron density in the doped semiconductor is _____ $\times 10^9/\text{m}^3$.
68. In the logic circuit shown in the figure, if input A and B are 0 to 1 respectively, the output at Y would be 'x'. The value of x is _____.

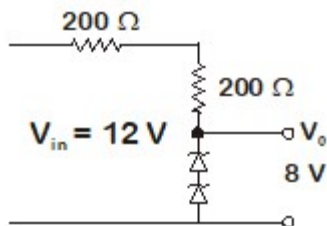


69. The typical output characteristics curve for a transistor working in the common-emitter configuration is shown in the figure.



The estimated current gain from the figure is _____.

70. The circuit shown below is working as a 8 V dc regulated voltage source. When 12 V is used as input, the power dissipated (in mW) in each diode is; (considering both zener diodes are identical) _____.



71. A particle of mass $200\text{ MeV}/c^2$ collides with a hydrogen atom at rest. Soon after the collision the particle comes to rest, and the atom recoils and goes to its first excited state. The initial kinetic energy of the particle (in eV) is $\frac{N}{4}$. The value of N is _____ .
(Given the mass of the hydrogen atom to be $1\text{ GeV}/c^2$)
72. A beam of electrons of energy E scatters from a target having atomic spacing of $1\text{ \AA}$. The first maximum intensity occurs at $\theta = 60^\circ$. Then E (in eV) is _____ .
(Planck constant $h = 6.64 \times 10^{-34}\text{ Js}$, $1\text{ eV} = 1.6 \times 10^{-19}\text{ J}$, electron mass $m = 9.1 \times 10^{-31}\text{ kg}$)



ANSWER KEY UNIT-5

1.	(2)	2.	(2)	3.	(4)	4.	(3)	5.	(3)
6.	(4)	7.	(3)	8.	(3)	9.	(3)	10.	(3)
11.	(4)	12.	(1)	13.	(2)	14.	(2)	15.	(1)
16.	(4)	17.	(1)	18.	(2)	19.	(4)	20.	(12)
21.	(4)	22.	(2)	23.	(3)	24.	(1)	25.	(1)
26.	(3)	27.	(2)	28.	(3)	29.	(3)	30.	(3)
31.	(2)	32.	(1)	33.	(None)	34.	(1)	35.	(2)
36.	(3)	37.	(2)	38.	(4)	39.	(4)	40.	(1)
41.	(3)	42.	(3)	43.	(4)	44.	(3)	45.	(2)
46.	(1)	47.	(3)	48.	(4)	49.	(2)	50.	(4)
51.	(1)	52.	(1)	53.	(4)	54.	(4)	55.	(2)
56.	(3)	57.	(3)	58.	(2)	59.	(3)	60.	(4)
61.	(198)	62.	(20)	63.	(05)	64.	(500)		
65.	(192)	66.	(25)	67.	(5)	68.	(0)		
69.	(200)	70.	(40)	71.	(51.00)	72.	(50.47)		